

tf.compat.v1.batch_scatter_update Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/batch_scatter_update

*Generalization of `tf.compat.v1.scatter_update` to axis different than 0.
(deprecated)*

Function Signatures

```
tf.compat.v1.batch_scatter_update( ref, indices, updates,  
use_locking=True, name=None )
```

Code Examples

```
tf.compat.v1.batch_scatter_update(  
ref, indices, updates, use_locking=True, name=None  
)
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Documentation

Generalization of `tf.compat.v1.scatter_update` to axis different than 0.
(deprecated)

Analogous to `batch_gather`. This assumes that `ref`, `indices` and `updates` have a series of leading dimensions that are the same for all of them, and the updates are performed on the last dimension of `indices`. In other words, the dimensions should be the following:

```
num_prefix_dims = indices.ndims - 1
batch_dim = num_prefix_dims + 1
updates.shape = indices.shape + var.shape[batch_dim:]
```

```
updates.shape[:num_prefix_dims]
== indices.shape[:num_prefix_dims]
== var.shape[:num_prefix_dims]
```

And the operation performed can be expressed as:

```
var[i_1, ..., i_n, indices[i_1, ..., i_n, j]] = updates[i_1, ..., i_n, j]
```

When `indices` is a 1D tensor, this operation is equivalent to `tf.compat.v1.scatter_update`.

To avoid this operation there would be 2 alternatives:

- 1) Reshaping the variable by merging the first `ndims` dimensions. However, this is not possible because `tf.reshape` returns a `Tensor`, which we cannot use `tf.compat.v1.scatter_update` on.
- 2) Looping over the first `ndims` of the variable and using `tf.compat.v1.scatter_update` on the subtensors that result of slicing the

first

dimension. This is a valid option for `ndims = 1`, but less efficient than this implementation.

See also `tf.compat.v1.scatter_update` and `tf.compat.v1.scatter_nd_update`.

Returns

Raises